

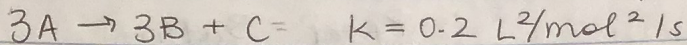
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11/29/2020

CHE 321 Homework #1

1. CSTR reactor, third-order, liq-phase rxn



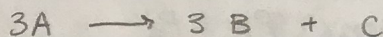
along w/ I

$$n_{\text{tot}} = 2400 \text{ mol/min}$$

$$X_{A0} = 0.75$$

$$V_0 = 100 \text{ L/min}$$

$$(a) \quad F_{A0} = (2400 \text{ mol/min})(0.75) = 1800 \text{ mol/min}$$



$$\text{Initial} \quad 1800 \quad 0 \quad 0$$

$$\text{Reaction} \quad -1800x \quad +1800x \quad +600x$$

$$\text{After} \quad 1800(1-x) \quad 1800x \quad 600x$$

(b) CSTR volume for $X = 0.8$

$$IN - OUT + GEN = 0$$

$$F_{A0} - F_A - r_A V = 0$$

$$F_{A0} = 1800 \text{ mol/min}$$

$$X = \frac{F_{A0} - F_A}{F_{A0}} \Rightarrow 0.8 = \frac{1800 \frac{\text{mol}}{\text{min}} - F_A}{1800 \frac{\text{mol}}{\text{min}}} \Rightarrow F_A = 360 \frac{\text{mol}}{\text{min}}$$

$$r_A = k C_A^3$$

$$r_A = (0.2 \text{ L}^2/\text{mol}^2/\text{s}) \left(C_A = \frac{F_A}{V} = \frac{360 \frac{\text{mol}}{\text{min}}}{100 \text{ L/min}} \right)^3 = 3.6 \frac{\text{mol}}{\text{L}}^3$$

$$r_A = (0.2 \text{ L}^2/\text{mol}^2/\text{s}) (46.656 \text{ mol}^3/\text{L}^3)$$

$$r_A = 9.3312 \frac{\text{mol}}{\text{L} \cdot \text{s}}$$

$$r_A = (9.3312 \frac{\text{mol}}{\text{L} \cdot \text{s}}) \left(\frac{60 \text{ s}}{1 \text{ min}} \right) = 559.872 \frac{\text{mol}}{\text{min} \cdot \text{L}} \quad [=] \quad \frac{\text{mol}}{\text{L} \cdot \text{s}}$$

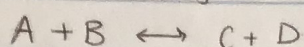
$$1800 \text{ mol/min} - 360 \text{ mol/min} - (559.872 \frac{\text{mol}}{\text{min} \cdot \text{L}})(V) = 0$$

$$-559.872 \frac{\text{mol}}{\text{min}} (V) = -1440 \text{ mol/min}$$

$$\frac{\text{mol}}{\text{min}} \cdot \frac{\text{L} \cdot \text{min}}{\text{mol}} [=] \text{ L}$$

$$V = 2.57 \text{ L}$$

2. reversible, gas-phase reaction, PFR



$$P_{\text{total}} = 3 \text{ atm}$$

$$C_{A0} = C_{B0} = 3 \text{ mol/L}$$

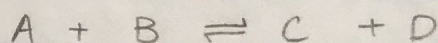
$$\text{fwd rxn rate } k_1 = 3 \text{ mol}/(\text{m}^3(\text{atm}^2)\text{s})$$

$$V_{0A} = V_{0B} = 0.5 \text{ L/s}$$

$$K = 1 \text{ (equilibrium constant)}$$

(a) max conversion (X) possible in this reaction?

$$r_A = 0 \text{ (no rxn) then } P_A = P_C$$



$$\text{Initial } P_{A0} \quad P_{A0} \quad 0 \quad 0$$

$$\text{Reaction } P_{A0}X \quad P_{A0}X \quad P_{A0}X \quad P_{A0}X$$

$$\text{After } P_{A0}(1-X) \quad P_{A0}(1-X) \quad P_{A0}X \quad P_{A0}X$$

$$\text{then } P_A = P_{A0}(1-X) = P_B$$

$$\text{and } P_C = P_D = P_{A0}X$$

$$\therefore P_A = P_C \Rightarrow P_{A0}(1-X) = P_{A0}X \Rightarrow 1-X = X$$

$$1 = 2X$$

$\therefore$ The max conversion

$$X = 1/2$$

possible for this reaction is 0.5

because $X \neq 1$ since the reaction is reversible

and $X \neq 0$ since the reaction must go forward as well.

(b) expression for conversion vs. volume of the reactor, and volume for the reactor will the conversion be 0.42?

find $V = f(X)$ then plug in $X = 0.42$

$$\boxed{\frac{dF_A}{dV} = r_A} \Rightarrow \int_0^V dV = \int_{F_{A0}}^{F_A} \frac{dF_A}{r_A} \Rightarrow \boxed{V = - \int_0^X \frac{F_{A0} dx}{r_A} = F_{A0} \int_0^X \frac{dx}{-r_A}}$$

$$F_{A0} = C_{A0} V_0 = (3 \text{ mols/L})(0.5 \text{ L/s}) = 1.5 \text{ mols/s}$$

$$r_A = -k_1 P_A P_B + k_{-1} P_C P_D$$

$$P_A = P_B; P_C = P_D$$

$$r_A = -k_1 P_A^2 + k_{-1} P_C^2$$

from part (a)

$$r_A = -k_1 [P_{A_0}(1-x)]^2 + k_{-1}(P_{A_0}x)^2$$

$$r_A = -(3) [1.5(1-x)]^2 + (3)(3x)^2$$

$$r_A = -3(1.5-1.5x)^2 + 27x^2$$

$$r_A = -3(1.5-1.5x)(1.5-1.5x) + 27x^2$$

$$r_A = -3(2.25 - 2.25x - 2.25x + 2.25x^2) + 27x^2$$

$$r_A = -6.75 + 13.5x - 6.75x^2 + 6.75x^2$$

$$r_A = -6.75 + 13.5x$$

$$V = \left(1.5 \frac{\text{mols}}{s}\right) \int_0^x \frac{1}{-(13.5x + 6.75)} dx$$

$$V = -\left(1.5 \frac{\text{mols}}{s}\right) \int_0^{0.42} \frac{1}{-13.5x - 6.75} dx$$

$$V = -\left(1.5 \text{ mols/s}\right) \int_{-6.75}^{-1.08} \frac{1}{u} du$$

$$V = -1.5 \text{ mols/s} \left(\frac{1}{13.5} \left(\ln |u| \right) \Big|_{-6.75}^{-1.08} \right)$$

$$V = -1.5 \text{ mols/s} \left(\frac{1}{13.5} \right) (\ln |-1.08| - \ln |-6.75|)$$

$$V = 0.20362 \text{ m}^3$$



$$1000 \text{ L} = 1 \text{ m}^3$$

$$V = 203.62 \text{ L}$$

$$u = 13.5x + 6.75 \quad \begin{matrix} x=0.42 & -6.75 \\ x=0 & -1.08 \end{matrix}$$

$$du = 13.5 dx$$

$$dx = \frac{du}{13.5}$$